

Northwind Traders — Event-Driven Order Platform on Azure

Production · West Europe · AKS microservices, Service Bus orders, event capture to the lake · fictional sample

PRODUCTION

- LEGEND
- HTTPS / app traffic
 - Private endpoint
 - Data flow
 - Sign-in (Entra ID)
 - Monitoring
 - Image pull

- KEY FLOWS
- Users → Front Door (HTTPS, WAF)
 - Front Door → App Gateway → AKS services
 - AKS services → Service Bus, Cosmos DB, SQL, Key Vault
 - Functions consume the orders queue, update Cosmos DB
 - Event Hubs Capture → Data Lake → Synapse
 - Workload telemetry → App Insights → Log Analytics

- DESIGN DECISIONS
- Private endpoints for every data service; public access disabled
 - Orders are commands on Service Bus; telemetry streams to Event Hubs
 - Event Hubs Capture lands raw events in the Data Lake for Synapse
 - AKS pods use workload identity; secrets live in Key Vault
 - Zone-redundant AKS, Cosmos DB, SQL and Service Bus Premium
 - Two WAF layers: Front Door at the edge, App Gateway in the VNet

